



SOUND — FULL MIND MAP

ध्वनि · Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway



Topics in this chapter:

1. Sound Propagation, Medium & Speed
2. Reflection, Echo, Reverberation & Acoustics
3. Pitch, Frequency, Wave Nature & Seismic Waves
4. Audible, Ultrasonic & Infrasonic Sound — Applications
5. Supersonic Motion & Mach Number
6. Decibel, Loudness & Noise Pollution
7. Sound Transmission, Musical Quality & Scale
8. Television Signal Transmission

1. Sound Propagation, Medium & Speed

- General exam order: gases < liquids < solids. Isliye Nitrogen < Water < Steel; air me sound ≈ 330 – 331 m/s aur metals/iron me isse kaafi faster hoti hai. [Q1–Q6]
- Sound ek mechanical wave hai: propagation ke liye particles wala material medium chahiye. Solid, liquid aur gas me travel kar sakti hai, lekin vacuum me nahi. [Q7–Q10]



EXAM TRAP — Q1 & Q6 — Speed of Sound: Density Alone Is Not the Rule

- ▶ Source explanations “higher density $\Rightarrow$ faster sound” ko simple rule ki tarah bolti hain. General physics me speed elasticity/stiffness aur density dono par depend karti hai (roughly $\propto \sqrt{\text{elastic modulus/density}}$).
- ▶ Exam-level comparison solid > liquid > gas usually correct hai, but sirf density ko universal cause mat samajhna.

2. Reflection, Echo, Reverberation & Acoustics

- Concert/music hall me excessive reflection ko control karne ke liye sound-absorbing/soft surfaces use ki jati hain; hard flat surfaces strong reflections aur echo create kar sakti hain. [Q11–Q12]
- Clear echo ke liye reflected sound ka delay ≈ 0.1 s chahiye. $d = vt/2$ se $v \approx 330$ – 340 m/s par reflector ki minimum distance ≈ 16.5 – 17 m aati hai. [Q13–Q14]
- Wave relation $v = f\lambda$. Q15 me $f = 4$ kHz aur $\lambda = 0.35$ m $\Rightarrow v = 1400$ m/s; 3 km travel time ≈ 2.14 s. [Q15]
- Stethoscope me heartbeat sound tube ki inner walls se multiple reflection ke through ears tak pahunchti hai. Large churches/buildings me repeated reflections se reverberation feel hota hai. [Q30, Q52]



EXAM TRAP — Q12–Q14 & Q52 — Echo vs Reverberation

- ▶ Echo me reflected sound original se sufficiently delayed hokar separately suna jata hai; reverberation me many reflections overlap karke sound ko prolonged banati hain.
- ▶ 17 m value $v \approx 340$ m/s aur persistence of hearing ≈ 0.1 s ke around estimate hai; exact value temperature ke saath thodi change ho sakti hai.

3. Pitch, Frequency, Wave Nature & Seismic Waves

- Pitch/shrillness ka primary determinant frequency hai: higher frequency $\Rightarrow$ higher pitch/shriller sound. Same medium me λ shorter hone par f higher hoti hai because $v = f\lambda$. [Q16–Q17]
- Moon ke surface par atmosphere nahi, practically vacuum hai; isliye astronauts direct sound wave se ek-dusre ko nahi sun sakte—radio communication use hota hai. [Q18–Q19]
- Shorter flute/air-column higher natural frequency produce karta hai. Gases/liquids me sound mainly longitudinal hoti hai; solids/rocks me longitudinal ke saath transverse elastic waves bhi propagate kar sakti hain. [Q20]
- Earthquake me P-waves faster hoti hain aur seismograph par pehle record hoti hain; P longitudinal/compressional, S transverse aur S-waves solids me hi propagate karti hain. [Q21]



EXAM TRAP — Q17 & Q20 — Frequency/Wavelength and Sound in Solids

- ▶ Shrillness directly frequency se decide hoti hai; wavelength ko direct determinant tabhi bolna safe hai jab propagation speed fixed ho.
- ▶ “Sound is always longitudinal” gas/liquid context me useful simplification hai. Solids me shear elasticity ke karan transverse elastic waves bhi possible hain.

4. Audible, Ultrasonic & Infrasonic Sound — Applications

- Human audible range ≈ 20 Hz– 20 kHz. Ultrasound > 20 kHz hota hai; infrasound < 20 Hz. Source ke 10 kHz threshold wale statements incorrect hain. [Q23–Q25]

- Ultrasonic waves cleaning, insect control, medical treatment/diagnosis, automatic sensing, submarine/depth detection jaise applications me use hoti hain. [Q22]
- Sonography/ultrasonography fetus ki growth aur pregnancy imaging ke liye ultrasonic waves use karti hai. [Q26–Q27]
- Bats ultrasonic pulses emit karke returning echoes se obstacle/prey ki position, direction aur size estimate karte hain—echolocation. [Q28]
- Q29 statement-set me source heart vibration ko infrasonic, Mach number ko speed-of-sound ratio aur ultrasound ko >20 kHz mark karta hai; gas me sound fastest wala statement false hai. [Q29]

⚠ EXAM TRAP — Q29 — “Heart Vibrates at Infrasonic” Is Source-Specific

- ▶ Cardiac vibration/sound spectrum me very-low-frequency components ho sakte hain, lekin stethoscope se sunai dene wali useful heart sounds audible band components bhi rakhti hain.
- ▶ Exam me Q29 ka intended combination source ke अनुसार preserve kiya gaya hai; universal statement ki tarah memorize na karein.

5. Supersonic Motion & Mach Number

- Mach number = object speed / local speed of sound. Mach >1 motion supersonic kehlata hai. [Q29, Q32]
- Source supersonic jet emissions (especially NO_x at high altitude) ko stratospheric ozone depletion potential se link karta hai. [Q31]
- Same Mach 1 par absolute aircraft speed local speed of sound ke equal hoti hai. Colder high-altitude air me sound speed lower hoti hai, so Mach 1 ka m/s value sea level se lower ho sakta hai. [Q32]

⚠ EXAM TRAP — Q32 — Mach 1 Is Not One Fixed km/h Value

- ▶ Mach number ratio hai, fixed speed nahi. Local temperature/medium change hone par speed of sound aur therefore Mach 1 ka actual speed change hota hai.

6. Decibel, Loudness & Noise Pollution

- Decibel (dB) sound level/noise level express karne ke liye use hota hai. Sound intensity level logarithmic scale par report hota hai. [Q33–Q35, Q43]
- Source reference values: normal conversation ≈30–60 dB, rustling leaves ≈20 dB, machine shop/instrument factory around 100 dB. [Q36–Q38]
- Source city-safe value 45 dB deta hai; continuous ≈80 dB+ exposure harmful batata hai, tolerable limit around 85–90 dB aur optimum/normal level around 60 dB mention karta hai. [Q39, Q41–Q44]
- India noise-rule matching: Residential 55 dB day, Silent zone 50 dB day, Commercial 65 dB day; Industrial standard 75 dB day / 70 dB night. Source list me 70 dB diya hone se industrial ko 70 se match kiya gaya. [Q40]
- Loudness perception amplitude/intensity se strongly related hai: larger amplitude = larger intensity = generally louder perceived sound. [Q45]

⚠ EXAM TRAP — Q39–Q40 — Noise Limits Depend on Context

- ▶ “Safe city noise = 45 dB” source-specific exam figure hai; modern health guidance often day/night, average exposure and source category ke hisab se different limits use karti hai.
- ▶ Q40 me industrial 70 dB source matching NIGHT limit se align hota hai; Indian rule ka DAY industrial limit 75 dB hai.

7. Sound Transmission, Musical Quality & Scale

- Steel rail me sound air se faster travel karta hai, isliye rail-track blow pehle rail ke through aur baad me air ke through sunai de sakta hai—two arrivals. [Q46]
- Solid → gas jane par sound speed generally decrease hoti hai. Air me temperature badhne se sound speed increase hoti hai; aluminium me sound speed glass se greater hoti hai. [Q47]
- Sound ek medium se doosre me enter kare to source-set frequency unchanged rehti hai; speed aur wavelength medium ke according change hote hain. [Q48]
- Sitarist string tension adjust karke natural frequency/pitch tune karta hai. Same pitch/loudness ke bawajood sitar aur flute ko quality/timbre se distinguish kiya ja sakta hai. [Q49–Q50]
- Linguistics me phoneme speech-sound ka basic contrastive unit hai. Physics ke sound-wave “basic unit” ke sense me ye term use nahi hota. [Q51]
- Source diatonic-scale item octave repetition ko include karke 8 tones count karta hai; first Sa=256 Hz ho to upper-octave Sa≈512 Hz aur source Ni≈480 Hz deta hai. [Q53]

 EXAM TRAP — Q51 & Q53 — Linguistics / Musical-Scale Wording

- ▶ Phoneme “basic unit of sound” language/phonology ka term hai, physical sound wave ki SI/basic unit nahi.
- ▶ Seven distinct scale notes/swaras ko octave Sa repeat karne par sequence me 8 tone positions dikhte hain; source Q53 isi counting convention ko use karta hai.

8. Television Signal Transmission

- Terrestrial television signals largely line-of-sight propagate karte hain; Earth curvature ki wajah se direct reception ek distance ke baad limited ho jati hai. [Q54]
- Source ke according modern TV me audio-video synchronized start hote hain; old television sets me picture warm-up delay ho sakta tha. [Q55]
- Conventional analog TV broadcasting me picture/video signal amplitude modulation (AM) aur sound/audio signal frequency modulation (FM) se transmit kiya jata tha. [Q56–Q57]

 EXAM TRAP — Q56–Q57 — Analog TV Context

- ▶ Picture-AM + audio-FM rule conventional analog television broadcasting ke liye hai. Modern digital TV systems different digital modulation/encoding schemes use karte hain.

✓ Source coverage: Q1–Q57